

Question 12.2

Find $f_{xx yzz}$ for $f(x, y, z) = z^3 y^2 \ln(x)$

(a) $12 \frac{zy}{x^2}$

(b) $-12 \frac{zy}{x^2}$

(c) $-6 \frac{z^2 y}{x^2}$

(d) $6 \frac{z^2 y}{x^2}$



Solution 12.2

Differentiating from left to right:

$$f_x = \frac{z^3 y^2}{x}$$

$$f_{xx} = -\frac{z^3 y^2}{x^2}$$

$$f_{xxy} = -\frac{2z^3 y}{x^2}$$

$$f_{xxyz} = -\frac{6z^2 y}{x^2}$$

$$f_{xxyz} = -\frac{12zy}{x^2}$$

Answer: (b)

